

Replication Plan

Motion Tomography via Occupation Kernels

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Auto-generated Replication Blueprint

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Paper Overview

This paper introduces a method for reconstructing spatially varying flow fields (e.g., ocean currents) from trajectory data of mobile sensors using *occupation kernels* in a reproducing kernel Hilbert space (RKHS) framework. The key idea is that a trajectory passing through a region “occupies” that region, and the occupation kernel captures the integrated effect of the flow field along the trajectory. The method solves a linear system relating trajectory-integrated measurements to the underlying vector field, without requiring spatial discretization (meshless). The authors validate on synthetic 2D flow fields with known ground truth.

Detailed Workflow

Phase 1: RKHS and Kernel Setup

1.1 Choose the base kernel function.

- Gaussian RBF kernel: $k(x, y) = \exp(-\|x - y\|^2 / (2\sigma^2))$.
- Exponential kernel: $k(x, y) = \exp(-\|x - y\| / \ell)$.
- Set kernel bandwidth σ or length scale ℓ .

1.2 Define the occupation kernel.

- For a trajectory $\gamma : [0, T] \rightarrow \mathbb{R}^d$, the occupation kernel is:

$$K_\gamma(x) = \int_0^T k(\gamma(t), x) dt$$

- Compute via numerical quadrature (Simpson’s rule or Gauss–Legendre).

1.3 Compute the Gram matrix of occupation kernels between all trajectory pairs:

$$G_{ij} = \int_0^{T_i} \int_0^{T_j} k(\gamma_i(t), \gamma_j(s)) ds dt$$

Phase 2: Synthetic Data Generation

2.1 Define ground-truth flow fields.

- Double gyre flow: $\psi(x, y) = A \sin(\pi f(x, t)) \sin(\pi y)$ with time-dependent parameters.
- Rankine vortex, uniform flow with perturbation, or other specified fields.

2.2 Generate synthetic trajectories.

- Integrate the ODE $\dot{x} = v(x, t)$ using RK4 from various initial conditions.
- Add measurement noise to trajectory points if specified.
- Vary number of trajectories N and trajectory duration T .

Phase 3: Flow Field Reconstruction

3.1 Assemble the linear system.

- The occupation kernel framework yields $\mathbf{G}\boldsymbol{\alpha} = \mathbf{b}$, where $\mathbf{b}$ encodes trajectory displacement data.

3.2 Solve the regularized linear system (Tikhonov regularization):

$$\boldsymbol{\alpha} = (\mathbf{G} + \lambda \mathbf{I})^{-1} \mathbf{b}$$

3.3 Evaluate the reconstructed field at a grid of query points:

$$\hat{v}(x) = \sum_i \alpha_i K_{\gamma_i}(x)$$

Phase 4: Error Analysis and Validation

4.1 Compute reconstruction error.

- RMSE between reconstructed and true flow field on a grid.
- Relative L^2 error norm.

4.2 Convergence studies.

- Error vs. number of trajectories.
- Error vs. kernel bandwidth.
- Error vs. regularization parameter λ .

4.3 Reproduce paper figures (flow field reconstructions, error maps, convergence plots).

Datasets

Dataset / Source	Type	Location / Notes
Synthetic flow fields	Generated	Double gyre, vortex, etc. from analytic expressions in the paper
Synthetic trajectories	Generated	ODE integration from specified initial conditions
Paper figure data	Reference	For quantitative validation of reconstruction quality

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
Python	≥ 3.8	Primary language
NumPy	≥ 1.21	Array operations, linear algebra
SciPy	≥ 1.7	ODE integration (<code>solve_ivp</code>), quadrature, linear system solvers
Matplotlib	≥ 3.4	Flow field visualization, quiver plots, contour maps
scikit-learn	≥ 0.24	Optional: kernel computations, Gaussian process utilities

Hardware Requirements

- Any modern laptop. Gram matrix is $N \times N$ (number of trajectories), typically $N \sim 100$ – 1000 .